

Examination Project

AI Engineer

Deadline: Mon, Oct 20, 2025

1 Project Objective

The primary objective of this project is to develop an **AI-based system** that automatically analyzes **Arabic patient feedback** collected from the hospital. The system will integrate advanced **Natural Language Processing (NLP)** techniques to extract meaningful insights, classify feedback, and support data-driven decision-making.

2 System Description

The proposed system will allow users to input patient feedback either as **text** or **speech**. The feedback text, written in Arabic, will be processed using the **GLiNER NLP model**, which leverages **Named Entity Recognition (NER)** to identify and extract relevant entities (e.g., symptoms, departments, services, or staff names) from the original text. These extracted entities will then be stored in their corresponding **database tables** for subsequent analysis and reporting.

3 Core Functionalities

1. Speech-to-Text (STT) Integration:

The system will enable voice input for patient feedback using STT models such as **Whisper**, **VOSK**, or similar tools. Model selection will be based on factors including **processing speed**, **accuracy**, **Arabic language support**, and **noise robustness** to ensure reliable transcription in hospital environments.

2. Feedback Understanding and Classification:

A machine learning module will analyze and categorize feedback into multiple classes (e.g., *medical*, *administrative*, *service quality*, etc.). This represents a **multi-class classification** problem, trained using labeled datasets to ensure accurate feedback categorization.

4 Operational Requirements

All system components, models, and libraries must operate in a **fully offline environment**, ensuring **data privacy**, **security**, and **compliance with hospital IT policies**.

5 Applicant Evaluation Criteria

Developers or contributors will be evaluated according to the following criteria:

- **Deadline Commitment:** Adherence to project timelines and deliverables.
- **Task Completion:** Fulfillment of assigned features and milestones.
- **Application Performance:** Efficiency, responsiveness, and robustness of the deployed system.
- **Code Quality:** Readability, modularity, and adherence to good programming practices (e.g., proper use of functions and classes).
- **System Stability:** Smooth, error-free, and consistent runtime performance in offline mode.